OPERATING SYSTEMS — COMPLETE CONCISE NOTES

PAGE 1 — OS FUNDAMENTALS & PROCESSES

1. Operating System

An OS is system software that manages hardware/resources and provides services to application programs. Main goals: convenience, efficient resource use, and protection.

2. OS Services & Components

- Program execution, I/O operations, file-system manipulation, communication, error detection, resource allocation, accounting and protection/security.
- Kernel: core of the OS. System call: controlled interface through which programs request kernel services.
- OS structures: monolithic, layered, microkernel, modular and hybrid approaches.

3. Processes

- Process = program in execution. PCB stores process state, program counter, registers, scheduling and memory information.
- States: New → Ready → Running → Waiting/Blocked → Terminated. Context switch saves one process state and loads another.
- Process creation uses parent/child relationships; termination may be normal or due to error/kill.

4. Threads

- Thread is the basic unit of CPU execution inside a process. Threads share code/data/files but have their own registers and stack.
- User-level threads are managed by a library; kernel threads are managed by the OS. Multithreading improves responsiveness and resource sharing.

5. CPU Scheduling

- Criteria: CPU utilization, throughput, turnaround time, waiting time, response time.
- FCFS: non-preemptive; simple but may cause convoy effect.
- SJF: shortest next CPU burst; minimizes average waiting time when burst lengths are known.
- SRTF: preemptive SJF.
- Priority: highest-priority process runs; starvation can be reduced by aging.
- Round Robin: each process receives a time quantum; suitable for time-sharing.
- Multilevel Queue / Multilevel Feedback Queue: processes are separated or allowed to move between priority queues.

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MEMORY MANAGEMENT

1. Address & Memory Concepts

- Logical/virtual address is generated by CPU; physical address is the actual RAM location.
- Address binding may occur at compile time, load time or execution time.
- Swapping moves processes between main memory and backing store.

2. Contiguous Allocation

- Fixed/variable partitions allocate contiguous memory to processes.
- Internal fragmentation: unused space inside an allocated block. External fragmentation: free space exists but is scattered.
- Placement strategies: first fit, best fit, worst fit.

3. Paging

- Logical memory is divided into fixed-size pages; physical memory into frames. Page table maps page number → frame number.
- TLB caches page-table entries to speed address translation.
- Advantages: eliminates external fragmentation; supports non-contiguous allocation. Page-table overhead is a disadvantage.

4. Segmentation

Memory is divided according to logical program segments such as code, data and stack. A logical address contains segment number and offset.

5. Virtual Memory

- Allows execution of processes larger than physical memory. Demand paging loads a page only when needed.
- Page fault occurs when a referenced page is not in memory. OS loads it from disk and updates the page table.
- Thrashing occurs when excessive page faults cause the system to spend more time paging than executing.

6. Page Replacement

- FIFO: replace oldest page; simple and can show Belady's anomaly.
- Optimal: replace page whose next use is farthest in the future; theoretical benchmark.
- LRU: replace page least recently used; approximates optimal using recent history.
- Second-Chance/Clock: FIFO with reference-bit information.

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SYNCHRONIZATION & DEADLOCKS

1. Process Synchronization

Race condition occurs when concurrent processes access shared data and the final result depends on execution order. Critical section is the code accessing shared data.

Requirements of a critical-section solution

- Mutual exclusion: at most one process is in its critical section.
- Progress: if no process is inside, selection of the next process cannot be postponed indefinitely.
- Bounded waiting: a process should not wait forever after requesting entry.

2. Synchronization Tools

- Mutex lock: binary ownership mechanism.
- Semaphore: integer synchronization variable accessed using wait(P) and signal(V). Binary and counting semaphores are common.
- Monitor: high-level construct combining shared data and procedures with implicit mutual exclusion.
- Classical problems: bounded buffer, readers-writers, dining philosophers, sleeping barber.

3. Deadlock

Deadlock is a state where processes wait indefinitely for resources held by one another.

Four necessary conditions

- Mutual exclusion; Hold and wait; No preemption; Circular wait.

4. Deadlock Handling

- Prevention: break at least one necessary condition.
- Avoidance: allocate resources only when the resulting state remains safe.
- Detection and recovery: allow deadlock, detect it, then terminate processes or preempt resources.
- Safe state: there exists a safe sequence in which every process can obtain its remaining resources and finish.

5. Banker's Algorithm

- Uses Available, Max, Allocation and Need matrices. Need = Max – Allocation.
- Safety algorithm repeatedly finds a process whose Need ≤ Work, assumes it finishes, then adds its Allocation to Work.
- If all processes can finish, the state is safe; otherwise it is unsafe.

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FILE SYSTEMS, DISK & STORAGE

1. Files

- File attributes: name, type, size, location, protection, timestamps and owner.
- Operations: create, open, read, write, seek, close, delete and truncate.
- Access methods: sequential, direct/random and indexed.

2. Directory Structures

- Single-level, two-level, tree-structured, acyclic-graph and general-graph directories.
- Path names may be absolute or relative.

3. File Allocation

- Contiguous allocation: fast sequential/direct access but suffers external fragmentation and file growth issues.
- Linked allocation: blocks can be scattered and files grow easily; direct access is inefficient.
- Indexed allocation: index block stores pointers to file blocks; supports direct access but needs index storage.

4. Free-Space Management

Common methods: bit vector/bitmap, linked list, grouping and counting.

5. Disk Scheduling

- FCFS: services requests in arrival order; fair but may have high head movement.
- SSTF: services nearest request; lowers movement but can starve distant requests.
- SCAN: elevator algorithm; head moves in one direction servicing requests, then reverses.
- C-SCAN: services in one direction only, then jumps back to the beginning.
- LOOK: like SCAN but reverses at the last pending request.
- C-LOOK: like C-SCAN but jumps from the last pending request to the first pending request.

6. RAID

- RAID combines disks for performance, redundancy or both.
- RAID 0: striping, no redundancy. RAID 1: mirroring. RAID 5: distributed parity. RAID 6: dual parity.

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PROTECTION, SECURITY & IPC

1. Protection

- Protection controls how processes/users access OS resources.
- Principle of least privilege: give only the permissions required.
- Protection domain defines the set of resources and operations available to a subject.
- Access matrix represents subjects/domains, objects and permitted operations.
- ACL associates permissions with an object; capability list associates accessible objects with a subject.

2. Security

- Authentication verifies identity; authorization determines permitted actions.
- Program threats: Trojan horse, trap door, buffer overflow and viruses.
- System/network threats include worms, denial of service and intrusion attempts.
- Security mechanisms include encryption, access control, authentication, auditing, firewalls and intrusion detection.

3. Interprocess Communication (IPC)

- Shared memory: processes communicate through a common memory region; synchronization is required.
- Message passing: processes exchange send/receive messages; can be direct/indirect and blocking/non-blocking.

4. Multiprocessor & Distributed Systems

- Multiprocessing uses multiple CPUs/cores. Benefits include higher throughput and improved reliability.
- Distributed systems coordinate multiple networked computers and may provide resource sharing and fault tolerance.

5. Virtualization

A virtual machine provides an abstraction of computer hardware. A hypervisor/VMM manages virtual machines and their access to physical resources.

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EXAM QUICK REVISION + IMPORTANT FORMULAS

CPU Scheduling

- Turnaround Time = Completion Time − Arrival Time.
- Waiting Time = Turnaround Time − Burst Time.
- Response Time = First CPU Start Time − Arrival Time.
- Know Gantt-chart calculation for FCFS, SJF, SRTF, Priority and Round Robin.

Page Replacement

- Page fault → trap to OS → locate page on disk → find free frame or select victim → load page → update page table → restart instruction.
- Practice FIFO, Optimal and LRU page-reference-string problems.

Deadlock

- Remember the four conditions: Mutual Exclusion + Hold and Wait + No Preemption + Circular Wait.
- Banker's Algorithm: Need = Max − Allocation; check Need ≤ Work; then Work = Work + Allocation.

Disk Scheduling

- For numerical problems, draw the disk request sequence and calculate total head movement.
- Know the movement pattern: FCFS → arrival order; SSTF → nearest; SCAN → elevator; C-SCAN → one direction; LOOK/C-LOOK → stop at last request.

High-Priority Exam Topics

- Process states and PCB; CPU scheduling numericals; threads; paging and page tables; page replacement numericals; virtual memory and thrashing.
- Semaphores and classical synchronization problems; deadlock conditions; Banker's Algorithm; deadlock detection/recovery.
- File allocation; directory structures; disk scheduling numericals; RAID; protection and security; IPC.

One-Line Memory Map

Process → **CPU Scheduling** → **Synchronization** → **Deadlock** → **Memory** → **Virtual Memory** → **File System** → **Disk** → **Protection** → **Security** → **IPC**

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